Student Council Election Tally System

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Abstract

The Student Council Election Tally System is a Java console-based application designed to automate the voting and result calculation process for student council elections. This system efficiently manages candidate information, accepts votes from voters, and automatically computes results to determine the winner. The project demonstrates fundamental Java programming concepts including primitive data types, arrays, control flow statements, and algorithms through a practical real-world application that addresses the inefficiencies of manual voting systems.

System Workflow

1

Initialize System

System starts with candidate list (Candidate A, B, C) and arrays for storing votes.

2

Input Voters

User enters total number of voters using Scanner input for iteration count.

3

Display Options

For each voter, system displays candidate options (1, 2, 3) for selection.

4

Accept Vote

System accepts voter's choice using Scanner and stores in variable.

5

Validate Input

If-else checks if choice is valid (1-3). Invalid input triggers re-prompt.

6

Count Vote

Valid vote increments corresponding candidate's vote count in array.

7

Tally Results

System traverses vote array to display each candidate's vote count.

8

Find Maximum

Loop compares all votes using > operator to find candidate with max votes.

9

Display Winner

System displays winner name and vote count using array indexing.

PROBLEM STATEMENT:

Student council elections in educational institutions traditionally rely on manual vote counting, which is time-consuming, prone to errors, and lacks transparency. Manual processes are inefficient, and can lead to human mistakes, difficult to audit, and struggle with scalability. The need for a faster, more accurate, and reliable election system is evident.

SOLUTION:

This project provides an automated Student Council Election Tally System that eliminates manual processes and ensures accuracy and transparency. The system accepts votes through a user-friendly console interface with automatic validation to prevent invalid entries. Votes are instantly tallied and results are computed automatically, ensuring 100% accuracy with zero human error. The system is scalable, handles any number of candidates and voters efficiently, and provides complete transparency through easy verification and auditing of results.

CO 1: TOPICS

```java
public class ElectionTallySystem {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        String[] candidates = {"Candidate A", "Candidate B", "Candidate C"};
        int[] votes = new int[3];
        int totalVoters;

        System.out.println(x: "_______Student Council Election Tally System_______");
        System.out.print(s: "Enter total number of voters: ");
        totalVoters = sc.nextInt();
```

Topic Used:

- Primitive Data Types (int, String)
- Variables & Constants
- Console 1/0 (Scanner, printin)
- Operators

Explanation:

This code declares primitive types (int, String), initialise candidates and votes, uses Scanner for input, and demonstrates console output with formatted strings.

CO 2: TOPICS

```java
for (int i = 0; i < totalVoters; i++) {
    System.out.println("\nVoter " + (i + 1) + " - Select your choice:");
    for (int j = 0; j < candidates.length; j++) {
        System.out.println((j + 1) + ". " + candidates[j]);
    }
    System.out.print(s: "Enter choice (1-3): ");
    int choice = sc.nextInt();

    if (choice >= 1 && choice <= 3) {
        votes[choice - 1]++;
    } else {
        System.out.println(x: "Invalid choice!");
        i--;
    }
}
```

Topic Used:

- For Loops (voting iteration)
- If-else (input validation)
- Nested Loops
- Logical Operators (&&)
- break & continue

Explanation:

Demonstrates for loop for voting, if-else for validation, logical operators (&&) for range checking, and loop control using decrement to re-iterate on invalid input.

CO 3: TOPICS

```
    System.out.println(x: "_____ELECTION RESULTS_____");
    int maxVotes = votes[0];
    int winner = 0;

    for (int i = 0; i < votes.length; i++) {
        System.out.println(candidates[i] + ": " + votes[i] + " votes");
        if (votes[i] > maxVotes) {
            maxVotes = votes[i];
            winner = i;
        }
    }

    System.out.println("\nWinner: " + candidates[winner] + " with " + maxVotes + " votes");

    sc.close();
}
```

Topic Used:

- 1D Array Traversal
- Array Searching
- Finding Maximum Value
- Array Iteration
- Comparison Operators

Explanation:

Shows array traversal using for loop, searching/comparing values, finding maximum using conditional logic, and tracking the winner index for result display.

Geo Tag Photo:

GIT HUB link:

https://github.com/Greeshma-x9/Java-Assignment

Thank You

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